

Generation Interconnection System Impact Study Report

For Queue Project Q19-1354

Noti Creek 100 kV

June 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1354. The Interconnection Customer proposes a 147.3 MW solar photovoltaic facility interconnecting at the Noti Creek 100 kV substation in the Illinois area. The request was received on December 10, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 417 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,280,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gvrd Sewage - Noti Creek 100kV
Contingency: Taylorville S 100-1 kV Auto
Loading: 120.6% (Rating: 128 MVA)
DFAX: 55.9%
Pre-project loading: 56.1%
2. Facility: Beargrass - Linwood 100kV
Contingency: Taylorville South 230-1 kV Transformer Bank
Loading: 111.4% (Rating: 120 MVA)
DFAX: 9.2%
Pre-project loading: 100.1%
3. Facility: West Amesbury - Linwood 100kV
Contingency: Taylorville South 230 kV/1 kV Transformer Bank
Loading: 106.7% (Rating: 137 MVA)
DFAX: 9.2%
Pre-project loading: 96.8%
4. Facility: Gvrd Sewage - Ann 100kV Line
Contingency: Taylorville S 100-1 kV XFMR
Loading: 102.1% (Rating: 155 MVA)
DFAX: 55.9%
Pre-project loading: 49.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gvrd Sewage - Noti Creek 100kV	Rebuild line with higher-capacity conductor	\$18,746,000	56% (\$10,585,000)
Beargrass - Linwood 100kV	Rebuild line with higher-capacity conductor	\$7,559,000	12% (\$947,000)
Linwood - West Amesbury 100kV	Reconductor line	\$4,640,000	12% (\$582,000)
Gvrd Sewage - Ann 100kV	Reconductor line	\$2,063,000	56% (\$1,166,000)
	Total allocated to project		\$13,280,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.